

The Problem with TF Alone

TF (Term Frequency) only counts how often a word appears in a document.

Suppose we have this document:

"The movie was good. The movie was really good."

Word	TF
the	2
movie	2
good	2
really	1

According to TF, **"the"** is just as important as **"good"**.

But is "the" useful for understanding the document?

No.

"The" appears in almost every English document.

The Problem with IDF Alone

Now imagine these documents:

D1: AI is amazing
D2: AI is the future
D3: AI is everywhere

Suppose another document contains:

AI AI AI AI AI

IDF tells us that:

- AI appears in every document.
- Therefore AI is not rare.
- So its IDF becomes very small.

But IDF ignores that **this document is mostly about AI**.

So IDF alone is also not enough.

TF-IDF Solves Both Problems

TF tells us

Is this word important **inside this document**?

IDF tells us

Is this word unique **across all documents**?

Multiply them together:

$$\text{TF-IDF} = \text{TF} \times \text{IDF}$$

Now a word gets a high score **only if**:

- it appears many times in this document (high TF)
 - it appears in very few other documents (high IDF)
-

Real Example

Imagine three documents.

D1:
AI AI AI Machine Learning

D2:
Football Match Goal

D3:
Football Player Goal

STEP 1 — TF

In D1

Word	TF
AI	3
Machine	1

Learning	1
----------	---

AI has the highest TF.

STEP 2 — IDF

Suppose

Word	Appears in Documen ts	IDF
AI	1	High
Machine	1	High
Learning	1	High
Football	2	Lower
Goal	2	Lower

STEP 3 — TF × IDF

Word	TF	IDF	TF-IDF
AI	3	High	Very High
Machine	1	High	High
Learning	1	High	High

Now it's obvious that **AI** is the keyword of Document 1.

Another Example

Suppose we have 1,000 documents.

The word "**computer**" appears

- 15 times in one document
- only 5 documents overall

Very important.

Now consider "**the**"

- appears 30 times
- appears in all 1,000 documents

Even though TF is larger, its IDF is almost zero.

computer

TF = 15

IDF = 5.3

TF-IDF = 79.5

the

TF = 30

IDF = 0.02

TF-IDF = 0.6

So **computer** is identified as much more informative than **the**.

Why Not Use Only TF?

TF cannot distinguish between meaningful words and very common words.

Why Not Use Only IDF?

IDF doesn't tell you whether a word is actually important in a particular document.

Why Multiply TF and IDF?

Multiplication keeps only words that satisfy **both** conditions:

- High frequency in the current document (high TF)
- Rare across the whole collection (high IDF)

If either condition is not met, the TF-IDF score stays low.

A Simple Analogy

Imagine you're looking for the topic of a book.

- **TF** asks: *"How many times is this word mentioned in this book?"*
- **IDF** asks: *"Is this word common in every book, or is it special to this one?"*

A word is a good keyword only if it is **mentioned frequently in this book** and **not common in most other books**. That's exactly what **TF-IDF** measures.

In one sentence: TF-IDF helps identify the words that best represent the content of a document by rewarding words that are frequent in that document but uncommon across the entire corpus.